

NTMS Terraform D-3

35 Questions

1. You have a Terraform project with multiple subdirectories: dev, staging, and prod. Each directory contains a separate Terraform configuration for each different environment. Before deploying resources, where do you need to run terraform init?

0/0 ☐ A Only in the root directory to initialize all environments at once.

0/0 ☐ B Only in prod since production requires initialization, but dev and staging can share the configuration.

0/0 ☒ C In each directory (dev, staging, and prod) since each is a separate working directory.

0/0 ☐ D In any one of the directories, then copy the .terraform folder to the other directories.

2. True or False? By default, the terraform destroy command will prompt the user for confirmation before proceeding.

0/0 ☒ T True

0/0 ☐ F False

3. Aside from traditional code reviews, which Terraform command provides an opportunity for team members to review each other's work before deployment?

0/0 ☐ A terraform output

0/0 ☒ B terraform plan

0/0 ☐ C terraform validate

4. Which Terraform command checks modules, attribute names, and value types to ensure the configuration is syntactically valid and internally consistent?

0/0 ☐ A terraform format

0/0 ☒ B terraform validate

0/0 ☐ C terraform show

5. You have split a large module into multiple .tf files and rearranged several resource blocks without changing any arguments or references. What impact should this have when running a terraform plan

- 0/0 ☐ A Multiple changes. Providers decide the order based on when blocks appear in the configuration, which may change the plan
- 0/0 ☐ B Minor changes. Terraform plans resources in the order files are read, so moving blocks to different .tf files can slightly reorder actions.
- 0/0 ☐ C Significant changes. Terraform tracks each resource's location within each .tf file, so moving blocks can trigger replacements.
- 0/0 ☒ D No changes. Block order doesn't affect the plan because Terraform parses all .tf files in a module together during execution.

6. After creating several new Terraform configurations, you want to quickly format the files without editing each one manually. How can you update all the files at once?

- 0/0 ☐ A terraform init
- 0/0 ☒ B terraform fmt
- 0/0 ☐ C terraform refresh
- 0/0 ☐ D terraform validate

7. What happens when a terraform apply command is executed?

- 0/0 ☐ A the backend is initialized and the working directory is prepped
- 0/0 ☐ B creates the execution plan for the deployment of resources
- 0/0 ☒ C applies the changes required in the target infrastructure in order to reach the desired configuration
- 0/0 ☐ D reconciles the state Terraform knows about with the real-world infrastructure

8. You have an existing Google Cloud Storage bucket that was created manually. You want to bring it under Terraform management using a modern config-driven approach, so you add the following configuration:

```
import {  
  to = google_storage_bucket.data_lake  
  id = "bk-existing-bucket"  
}  
  
resource "google_storage_bucket" "data_lake" {  
  name    = "bk-existing-bucket"  
  location = "US"  
}
```

- 0/0 ☒ A run terraform apply -import-only to import without other changes
- 0/0 ☐ B run terraform init followed by terraform import
- 0/0 ☒ C run terraform plan followed by terraform apply to import the resource
- 0/0 ☐ D run terraform import google_storage_bucket.data_lake bk-existing-bucket

9. Which of the following actions are performed during a terraform init? (select three)

- 0/0 ☒ A initializes the backend configuration
- 0/0 ☒ B downloads the providers/plugins required to execute the configuration
- 0/0 ☐ C provisions the declared resources in your configuration
- 0/0 ☒ D downloads the required modules referenced in the configuration

10. Your team wants to enforce consistent formatting across all Terraform files before merging code into the main branch. You're setting up a CI/CD pipeline and need a command that checks whether files are properly formatted without making changes. Which command should you use?

- 0/0 ☒ A terraform fmt -check
- 0/0 ☐ B terraform fmt -validate
- 0/0 ☐ C terraform fmt -diff
- 0/0 ☐ D terraform validate -format

11. Your organization manages a Google Cloud project with 50 resources across multiple services. You need to decommission only the Cloud SQL database instance and its backup policy while keeping all other infrastructure running. What is the most appropriate approach?
- 0/0 ☒ A Add two removed blocks to your configuration, pointing to each of the Cloud SQL database and the backup policy
- 0/0 ☐ B Remove the database and backup resource blocks from your configuration, then run terraform apply.
- 0/0 ☐ C Run terraform destroy to remove all resources, then run terraform apply to recreate everything except the database
- 0/0 ☐ D Run terraform destroy -[target=google_sql_database_instance.main](#) -
[target=google_sql_backup_run.policy](#)
12. What is the purpose of Terraform's dependency graph?
- 0/0 ☐ A lets providers decide ordering, relying on cloud APIs to handle dependencies
- 0/0 ☒ B builds a graph of dependencies to perform create/update/destroy operations from it
- 0/0 ☐ C sorts resources by type and name and builds a visual graph from them for create operations
- 0/0 ☐ D executes resources in the order .tf files appear, ensuring predictable runs
13. Your team's change management process requires that all Terraform changes be reviewed and approved before execution. You run terraform plan -out=bk-[project.tfplan](#) and send the output for review. After approval is granted two hours later, what command should you run to execute the exact changes that were reviewed?
- 0/0 ☐ A run `terraform apply` to apply the latest configuration changes since Terraform automatically executes the last plan
- 0/0 ☒ B run `terraform apply bk-[project.tfplan](#)` to execute the saved plan
- 0/0 ☐ C run `terraform apply -auto-approve` to skip the confirmation prompt
- 0/0 ☐ D run `terraform plan` again to verify no drift occurred, then run `terraform apply`
14. Which Terraform command will force a resource to be destroyed and recreated even if there are no configuration changes that would require it?
- 0/0 ☐ A terraform apply -refresh-only
- 0/0 ☐ B terraform destroy
- 0/0 ☒ C terraform apply -replace=<address>
- 0/0 ☐ D terraform fmt

15. What is the [.terraform.lock.hcl](#) file and when does Terraform create or modify it?

- 0/0 ☐ A The [.terraform.lock.hcl](#) file is a concurrency lock to prevent simultaneous runs. It is created or updated every time you run a terraform plan
- 0/0 ☐ B The [.terraform.lock.hcl](#) file contains backend connection and credentials settings. It is created or updated every time you modify the backend
- 0/0 ☒ C The [.terraform.lock.hcl](#) file is a dependency lock file used by Terraform. It is created or updated every time you run terraform init.
- 0/0 ☐ D The [.terraform.lock.hcl](#) file is a state snapshot used by Terraform. It is created or updated every time you run terraform apply

16. After executing a terraform plan in your working directory, you notice that a resource has a tilde (~) next to it. What does this indicate?

- 0/0 ☒ A the resource will be updated in place
- 0/0 ☐ B Terraform can't determine how to proceed due to a problem with the state file
- 0/0 ☐ C the resource will be destroyed and recreated
- 0/0 ☐ D the resource will be created

17. What are the three core steps that make up the Terraform workflow? (select three)

- 0/0 ☒ A Write - Author infrastructure as code.
- 0/0 ☒ B Apply - Provision reproducible infrastructure.
- 0/0 ☐ C Validate - Verify the syntax and internal consistency
- 0/0 ☐ D Destroy - Deprovision and remove all infrastructure resources
- 0/0 ☒ E Plan - Preview changes before applying.

18. You're refactoring a large Terraform configuration, splitting a monolithic file into multiple smaller files and reorganizing resource blocks. Before running terraform plan, what's the fastest way to verify you didn't introduce any syntax errors during the refactoring?

- 0/0 ☐ A run **terraform plan**
- 0/0 ☒ B run **terraform validate**
- 0/0 ☐ C run **terraform fmt -recursive**
- 0/0 ☐ D run **terraform init**

19. You've completed writing your Terraform configuration and run `terraform plan`. The output shows **+/- 4 resources**, indicating that some resources are being destroyed and recreated. Before you run `terraform apply`, which actions should you take? (select two)

- 0/0 ☒ A check if the resources being replaced will cause downtime or data loss
- 0/0 ☐ B run **terraform apply** since the plan has already validated the configuration
- 0/0 ☒ C review which resources are being replaced and understand why Terraform is forcing replacement
- 0/0 ☐ D run **terraform init -upgrade** to ensure you have the latest provider versions
- 0/0 ☐ E run **terraform validate** to confirm the plan is correct

20. What function does the **terraform init -upgrade** command perform?

- 0/0 ☒ A update all previously installed plugins and modules to the newest version that complies with the configuration's version constraints
- 0/0 ☐ B upgrades the backend to the latest supported version
- 0/0 ☐ C upgrades the Terraform configuration file(s) to use the referenced Terraform version
- 0/0 ☐ D upgrades all of the referenced modules and providers to the latest version of Terraform

21. You have three test VMs running on a new cluster, and you run **terraform destroy** to remove them. However, Terraform only removes two of the virtual machines, leaving one virtual machine still running.

Why would Terraform only remove two of the three virtual machines?

- 0/0 ☐ A The vSphere provider credentials are invalid, and therefore, Terraform cannot reach the third virtual machine
- 0/0 ☐ B Terraform can only delete a maximum of 2 resources per **terraform destroy** run.
- 0/0 ☒ C The remaining virtual machine was not created by Terraform; therefore, Terraform is unaware of it and cannot destroy it.
- 0/0 ☐ D The last virtual machine was tagged with vSphere tags to prevent it from being deleted.

22. True or False? Terraform requires you to run a **terraform plan** before applying the configuration with **terraform apply**.

- 0/0 ☐ T True
- 0/0 ☒ F False

- 23.** You refactored a module and renamed a resource as shown in the code example below. When running terraform plan, Terraform wants to destroy [aws_s3_bucket.logs](#) and create aws_s3_bucket.app_logs. How can you make this a config-driven change so Terraform will update state to the new address without replacing it?

```
# ORIGINAL
resource "aws_s3_bucket" "logs" {}

# UPDATED
resource "aws_s3_bucket" "app_logs" {}
```

- 0/0** **A** add an **import** block with the **to** and **id** arguments in your configuration to indicate the change to Terraform
- 0/0** **B** run a **terraform apply** -replace="[aws_s3_bucket.logs](#)" before running **terraform apply** again
- 0/0** **C** remove the old resource from the configuration file before running **terraform apply** again
- 0/0** **D** add a **moved** block to indicate the address change directly in your configuration

- 24.** What built-in tool or command can you use to easily format Terraform code to meet the recommended canonical formatting and style?

- 0/0** **A** terraform fmt
- 0/0** **B** terraform validate
- 0/0** **C** terraform refresh

- 25.** A developer writes a new Terraform configuration to create a virtual machine. They run terraform validate and it reports "Success! The configuration is valid." However, when the developer runs terraform apply, the command fails with an error stating that the instance_type does not exist. Why did terraform validate not detect this error?

- 0/0** **A** The developer's API credentials were not configured, so validate skipped the provider check.
- 0/0** **B** The developer must run terraform plan first, as validate only checks variable definitions.
- 0/0** **C** The terraform validate command requires an -auto-approve flag to check for provider-level errors.
- 0/0** **D** The terraform validate command checks for syntactic correctness but does not communicate with the provider's API to validate resource-specific values.

- 26.** You have a Terraform workspace managing Azure infrastructure as shown in the exhibit below. You run `terraform destroy`. In what order will Terraform destroy these resources?

```
resource "azurerm_resource_group" "main" {
  name     = "production-bk"
  location = "East US"
}

resource "azurerm_virtual_network" "main" {
  name                = "prod-bk-vnet"
  resource_group_name = azurerm_resource_group.main.name
  location            = azurerm_resource_group.main.location
  address_space       = ["172.16.0.0/16"]
}

resource "azurerm_subnet" "main" {
  name                = "prod-bk-subnet"
  resource_group_name = azurerm_resource_group.main.name
  virtual_network_name = azurerm_virtual_network.main.name
  address_prefixes     = ["172.16.67.0/24"]
}
```

- 0/0** **A** Terraform will follow the dependency chain: subnet, virtual network, resource group
- 0/0** **B** Terraform will destroy them alphabetically by resource type: virtual network, subnet, resource group
- 0/0** **C** Terraform will follow the order they appear in the configuration: resource group, virtual network, subnet
- 0/0** **D** Terraform will destroy all three resources simultaneously due to Terraform's default parallelism configuration
- 27.** A developer has a valid Terraform configuration and has already successfully initialized the working directory. Assuming the developer approves the execution plan, which of the following actions will the `terraform apply` command perform? (select two)
- 0/0** **A** Update the state file to match the changes made to the infrastructure during the execution.
- 0/0** **B** Create, update, or delete infrastructure in the target environment based on the changes specified in your Terraform files.
- 0/0** **C** Downloads and installs the required provider plugins and modules referenced in the configuration.
- 0/0** **D** Terraform recreates all the infrastructure defined in the configuration file

28. You've updated a production load balancer using Terraform. Before making changes, you must (a) preview the exact set of actions as a dry run and (b) ensure that the same reviewed actions are applied later without drift. Which commands meet both requirements?
- 0/0 ☐ A terraform plan -out=lb.tfplan → terraform apply -auto-approve
- 0/0 ☐ B terraform fmt -check → terraform plan -refresh-only
- 0/0 ☒ C terraform plan -out=lb.tfplan → terraform apply lb.tfplan
- 0/0 ☐ D terraform validate → terraform apply
29. When running the terraform validate command, which issue will be brought to your attention?
- 0/0 ☐ A there is no existing state file for the configuration
- 0/0 ☐ B there is configuration drift within the managed infrastructure
- 0/0 ☐ C parameters inside of a resource block are not lined up with spaces
- 0/0 ☒ D a variable is being used in a resource block but has not been declared
30. You're working on a Terraform project deploying VMware workloads. You now need to add a public DNS record in Amazon Route53, so you add an aws provider block to your configuration file. After saving the file, what should you do next?
- 0/0 ☐ A run terraform validate to ensure the provider syntax is correct
- 0/0 ☐ B run terraform plan to validate the new DNS record will be created as needed
- 0/0 ☒ C run terraform init to download the newly required AWS provider
- 0/0 ☐ D run terraform apply since providers are downloaded automatically during apply
31. After running terraform init in a new working directory, Pooja wants a dry run that saves the result for a later apply without creating resources. You run **terraform plan -out=thomas**
- 0/0 ☒ T True
- 0/0 ☐ F False
32. You clone a Terraform repo, adjust a variable, and immediately run terraform apply. The command fails before planning, complaining about missing provider plugins. Which core workflow step should have happened first?
- 0/0 ☒ A Run terraform init to set up the working directory and install providers.
- 0/0 ☐ B Run terraform fmt to standardize files so planning can continue.
- 0/0 ☐ C Run terraform graph to review dependencies before any planning.

33. You must destroy and recreate only one database server managed by Terraform without editing the configuration. Which command should you run?

- 0/0 ☐ A terraform plan -destroy="aws_instance.database"
- 0/0 ☐ B terraform apply -target="aws_instance.database"
- 0/0 ☐ C terraform destroy -target="aws_instance.database"
- 0/0 ☒ D terraform apply -replace="aws_instance.database"

34. Which CLI commands will completely tear down and delete all resources that Terraform manages in the current workspace? (select two)

- 0/0 ☒ A terraform destroy
- 0/0 ☐ B terraform plan -destroy
- 0/0 ☒ C terraform apply -destroy
- 0/0 ☐ D terraform apply -delete

35. Your team has a CI/CD pipeline that runs terraform fmt -check, then terraform init, terraform validate, and finally terraform plan. A developer commits code with a typo in an argument name and writes instance_typo instead of instance_type. At which stage will the pipeline catch this error?

- 0/0 ☐ A terraform fmt -check
- 0/0 ☐ B terraform init
- 0/0 ☐ C terraform plan
- 0/0 ☒ D terraform validate